



Dual-Domain Balanced Channel-Spatial Mixed Attention in Convolutional Neural Networks

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Abstract

Channel-Spatial mixed attention methods have been proven to enhance the representation capability of convolutional neural networks. Most existing channel-spatial mixed attention methods decouple the channel domain and spatial domain, investing different orders of computational complexity in calculating attention for the two domains. However, both the channel domain and spatial domain are equally important and deserve comparable levels of complexity in attention computation. To solve this problem, we propose a novel channel-spatial mixed attention method, which is called **Dual-domain Balanced Channel-Spatial Mixed Attention (DBCSMA)**. The input tensor is first decomposed into channel and spatial domains using global pooling. Then, the same order of complexity is dedicated to calculating attention in both the channel domain and the spatial domain, and a mixed attention tensor is generated using multiplicative aggregation. Finally, input channel information and spatial information are simultaneously recalibrated using multiplication of this mixed attention tensor and the input tensor. Experimental results demonstrate that DBCSMA achieves superior performance compared to most existing channel-spatial mixed attention methods while using lower complexity.

Keywords Convolutional neural networks · Channel-Spatial mixed attention · Channel correlations · Spatial correlations

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1 Introduction

Convolutional neural networks (CNNs) have shown remarkable success across diverse tasks [1–3]. These networks utilize a series of filters to identify spatial relationships between adjacent input channels within each convolutional layer. By means of integrating position-specific and channel-specific information within confined receptive fields, CNNs are able to produce resilient representations. By means of stacking numerous convolutional layers, layers with nonlinear activation functions, and downsampling operators, CNNs efficiently identify hierarchical patterns, attaining a comprehensive theoretical receptive field.

Let the training sample come from $X \in \mathbb{R}^{C \times H \times W}$. Fig. 1 illustrates channel attention, spatial attention, and channel-spatial mixed attention. In this study, “channel” defines the high-dimensional representation of a feature, like three color channels an RGB image. “Spatial” defines positional indices of a feature, like positional indices of a color channel.

In recent times, various channel-spatial mixed attention techniques have showcased their capability to enhance the performance of CNNs [4–7]. Channel-Spatial mixed attention is aimed at adaptively recalibrated both channel information and spatial information to focus on the relevant objects and areas of interest. Most of the current channel-spatial mixed attention methods decouple the channel domain and spatial domain, investing different orders of computational complexity in calculating attention for the two domains. This issue can be substantiated by the experimental results listed in Table 1. However, both the channel domain and spatial domain are equally important and deserve comparable levels of complexity in attention computation.

To solve this problem, this paper proposes a novel channel-spatial mixed attention method called as **Dual-domain Balanced Channel-Spatial Mixed Attention (DBCSMA)**, as shown in Fig. 2. DBCSMA consists of two parallel gated paths: channel attention gated path and spatial attention gated path. These two gated paths are aggregated multiplicatively and the complexity of them is approximately equal. The channel attention gated path utilizes an inverted bottleneck to build channel correlations and generates a normalized channel attention tensor. The spatial attention gated paths utilizes a convolution layer to build spatial correlations and generates a normalized spatial attention tensor.

The primary contributions of this paper are summarized as follows.

- 1) We propose DBCSMA, a novel dual-branch channel-spatial mixed attention method aimed at establishing both channel correlations and spatial correlations to enhance representation. The core innovation lies in the unique design of two parallel gated paths: the channel attention gated path and the spatial attention gated path. This dual-branch structure allows for the simultaneous capture of channel-wise and spatial features, with

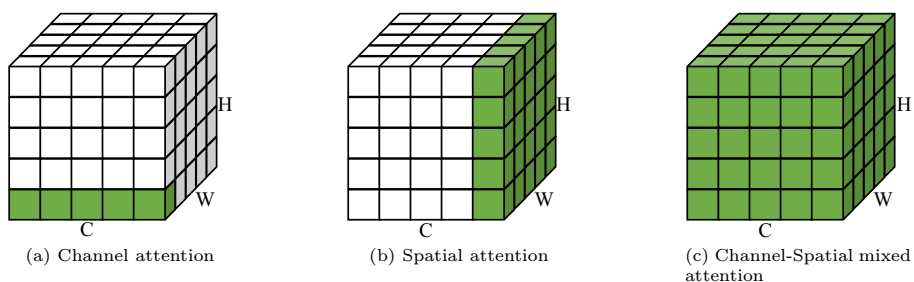


Fig. 1 Categories of attention

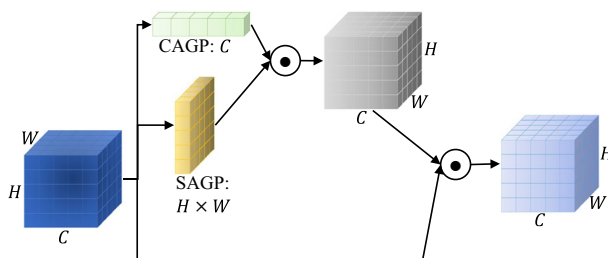
Table 1 Parameter complexity of the current channel-spatial mixed attention methods varies across different domains

Method	C	$H \times W$	$C \times H$	$C \times W$	$C \times H \times W$
CBAM [4]	$2C^2r$	$2k^2$	N/A	N/A	N/A
BAM [5]	$2C^2r$	$C^2r + 2C^2r^2k^2 + Cr$	N/A	N/A	N/A
FAB [8]	N/A	N/A	N/A	N/A	$2C^2r$
GALA [9]	$2C^2r$	$C^2r + Cr$	N/A	N/A	N/A
TA [10]	N/A	$2k^2$	$2k^2$	$2k^2$	N/A
CA [6]	N/A	N/A	$2C^2r$	$2C^2r$	N/A
SA [11]	C/G	N/A	N/A	N/A	$2C/G$
SimAM [12]	N/A	N/A	N/A	N/A	0
EMA [7]	N/A	$2C/G + k^2C^2/G^2$	C^2/G^2	C^2/G^2	N/A

Note 1: C denotes the number of channels, r denotes reduction ratio, and k denotes kernel size.

Note 2: N/A stands for Not Applicable.

Note 3: The parameter complexity of CA and EMA in the $C \times H$ and $C \times W$ domains are shared

**Fig. 2** Overview of DBCSMA. Note: CAGP stands for channel attention gated path, and SAGP stands for spatial attention gated path

approximately equal complexity invested in each path. A mixed attention tensor is generated using multiplicative aggregation, which effectively integrates the strengths of both channel and spatial attention.

- To validate the effectiveness of DBCSMA, we conducted comparative experiments on the CIFAR-100 [13] dataset and the PASCAL VOC 2007/2012 [14] dataset. The results demonstrate that DBCSMA outperforms most of the current channel-spatial mixed attention methods while using lower complexity. This highlights the efficiency and superiority of our proposed DBCSMA method.

The structure of the remainder of this paper is as follows. Sec. 2 offers an overview of the relevant literature. Our attention method is dissected in Sec. 3. The experimental results are presented in Sec. 4. The conclusion is presented in Sec. 5.

2 Related work

Channel-spatial mixed attention, which combines channel attention and spatial attention, enables the adaptive selection of important objects and regions [15]. The notion of channel-spatial attention was first introduced in the residual attention network [16], highlighting

the importance of informative features across both channel and spatial dimensions. This method utilized a bottom-up architecture, incorporating multiple convolutional layers to produce a three-dimensional attention tensor encompassing the channel, height, and width axes. However, it was plagued by substantial computational expenses and restricted receptive fields. Subsequent research [4–7, 9, 11] has improved the discriminative power of features and incorporated global spatial information. This has been achieved by integrating global average pooling (GAP) and decoupling channel attention from spatial attention, leading to enhanced computational efficiency.

Wang *et al.* [8] put forward the fully attentional block (FAB), which retained spatial information by eliminating the pooling layer. Yang *et al.* [12] emphasized the importance of obtaining attention weights that vary across both channel and spatial dimensions. They proposed SimAM, a straightforward and parameter-free attention module capable of directly computing three-dimensional weights, thereby obviating the requirement to expand one- or two-dimensional weights. Inspired by neuroscience theories, the design of SimAM obviates the requirement for manual refinement of the network architecture.

Woo *et al.* [4] proposed the convolutional block attention module (CBAM), which introduced a stacked series of channel attention and spatial attention mechanisms to emphasize informative channels and significant regions. To improve computational efficiency, the channel attention map and spatial attention map were separated, and global pooling was utilized to harness spatial global information. By sequentially combining channel attention and spatial attention, CBAM effectively harnessed both channel and spatial feature correlations, directing the network's attention to what and where it should focus. It prioritized beneficial channels and augmented informative localized regions. However, there is still room for enhancement in CBAM. *E.g.*, employing a convolutional layer to generate the spatial attention map may result in a constrained receptive field for the spatial sub-module. In response to this limitation, Park *et al.* [5] proposed the bottleneck attention module (BAM) alongside CBAM, aiming to enhance the network's representational capability while maintaining computational efficiency. BAM utilized dilated convolutions to broaden the receptive field of the spatial attention sub-module and incorporated a bottleneck architecture to decrease computational complexity. It concurrently inferred channel attention and spatial attention in separate streams, and then combined the two attention maps after resizing the outputs of both branches. Despite the dilated convolutions' effectiveness in enlarging the receptive field, they still encountered difficulties in capturing long-range positional information and encoding long-range positional correlations. The achievement of attention mechanisms that depend on weak supervisory signals derived from class labels motivated Linsley *et al.* [9] to investigate how explicit human supervision could enhance the performance and interpretability of attention models. As a demonstration of the idea's feasibility, they proposed the global-and-local attention (GALA) module, which augmented the SE block with an additional spatial attention mechanism. To tackle computational complexity, Zhang *et al.* [11] presented an efficient Shuffle Attention (SA) module that integrates two kinds of attention mechanisms through the use of Shuffle units. The SA module divided the channel dimensions into several sub-features and processed them simultaneously. For each sub-feature, a Shuffle unit was employed to capture the inter-dependencies among features in both the channel and spatial dimensions. The resultant sub-features were then combined, and a "channel shuffle" operation was used to enable the exchange of information among different sub-features.

In CBAM and BAM, the computations for channel attention and spatial attention were carried out independently, neglecting the interactions between these two domains [10]. To address this limitation and capture cross-dimension interactions, Misra *et al.* [10] presented the triplet attention (TA), an efficient and lightweight attention mechanism. Unlike CBAM

and BAM, TA emphasized capturing cross-domain interactions instead of independently computing spatial and channel attention. This approach allowed TA to capture more informative and discriminative feature representations.

While the SE block adeptly employed global average pooling to capture channel correlations through the aggregation of global spatial information, it overlooked the significance of positional information. On the other hand, BAM and CBAM were able to capture local position correlations through the use of convolutions, but encountered difficulties in modeling long-range position dependencies. To address these shortcomings, Hou *et al.* [6] presented a new attention method named as coordinate attention (CA). CA seamlessly integrated positional information into channel attention, enabling the network to pinpoint crucial regions. The CA process comprised two primary steps: embedding coordinate information and creating coordinate attention. Simultaneously, Ouyang *et al.* [7] proposed an ingenious and efficient multi-scale attention (EMA) module, designed to preserve channel-specific information while minimizing computational complexity. EMA divided the channel dimensions into multiple sub-features, guaranteeing an even spread of spatial semantic features among each feature group. Besides leveraging global information to adjust channel-wise weights in parallel branches, EMA incorporated a cross-dimensional interaction technique to smoothly integrate the output features of the two branches.

3 Proposed channel-spatial mixed attention method

This section first gives an overview of the method, then analyzes the gated path of channel attention, and finally analyzes the gated path of spatial attention.

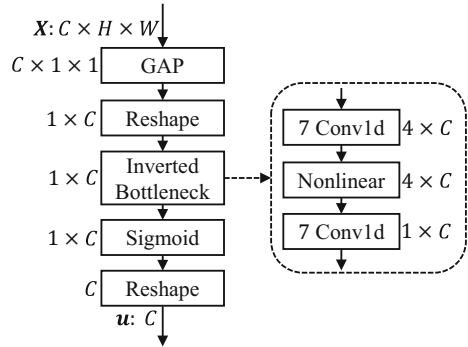
3.1 Overview of DBCSMA method

This section proposes DBCSMA, whose overview is illustrated in Fig. 2. DBCSMA consists of two parallel gated paths: the channel attention gated path (CAGP) and the spatial attention gated path (SAGP). The channel attention gated path is responsible for establishing channel correlations and outputting a channel attention tensor. The spatial attention gated path is responsible for establishing positional correlations and outputting a spatial attention tensor. Assuming that the input is $X \in \mathbb{R}^{C \times H \times W}$, the output of the channel attention gated path is $u \in \mathbb{R}^C$, the output of the spatial attention gated path is $v \in \mathbb{R}^{H \times W}$, and the output of DBCSMA is $Y \in \mathbb{R}^{C \times H \times W}$. The computation order of the overall framework can be described as Eq. (1),

$$\begin{aligned} \forall k &\in \{1, 2, \dots, C\}, \\ h &\in \{1, 2, \dots, H\}, \\ w &\in \{1, 2, \dots, W\}, \\ y_{k,h,w} &= (u_k v_{h,w}) x_{k,h,w} \end{aligned} \quad (1)$$

Prior knowledge suggests that the channel domain and spatial domain should have equal importance. Based on this, DBCSMA is designed to have approximately equal complexity for the two gated paths. As can be seen from Fig. 3, the parameter complexity required for the channel attention gated path is 56. As shown in Fig. 4, the parameter complexity required for the spatial attention gated path is 49. Therefore, the parameter complexity invested in the two gated paths is approximately equal.

Fig. 3 Channel attention gated path. Note: GAP stands for global average pooling



Addition is much faster than multiplication, so the computational complexity can be estimated by calculating the number of multiplications required. In neural network models, the dominant operation in terms of computation is matrix operations. By definition, multiplying a matrix of size $a \times b$ by a matrix of size $b \times c$ requires abc multiplications, which allows us to estimate the computational complexity of multiplying two matrices as abc . According to this computational estimation rule, the computational complexity required for the channel attention gated path is $\mathcal{O}(56C)$, and the computational complexity required for the spatial attention gated path is $\mathcal{O}(49HW)$, which can also be considered to be in the same order of magnitude. Therefore, the computational complexity invested in the two gated paths is also approximately equal.

3.2 Channel attention gated path

Fig. 3 plots the channel attention gated path.

Firstly, the channel attention gated path uses global average pooling to aggregate global spatial information and generate channel-wise statistics $\mu \in \mathbb{R}^C$, as shown in Eq. (2):

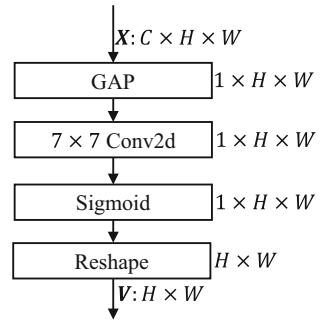
$$\mu_k = \frac{1}{HW} \sum_{h=1}^H \sum_{w=1}^W x_{k,h,w} \quad (2)$$

Next, the channel attention gated path establishes channel correlations using an Inverted Bottleneck. The Inverted Bottleneck consists of two convolutional layers and a nonlinear layer, as shown in Eq. (3):

$$z = \text{Conv}_7(\phi(\text{Conv}_7(\mu, \Theta_1)), \Theta_2) \quad (3)$$

where $z \in \mathbb{R}^C$, and $\phi(\cdot)$ denotes a nonlinear activation function. Assuming that the weight of the first convolutional layer is $\Theta_1 \in \mathbb{R}^{4 \times 1 \times 7}$ (4 denotes the number of output channels, 1 denotes the number of input channels, and 7 denotes the kernel size), and the weight of the second convolutional layer is $\Theta_2 \in \mathbb{R}^{1 \times 4 \times 7}$ (1 denotes the number of output channels, 4 denotes the number of input channels, and 7 denotes the kernel size). Then, Eq. (3) can be further expanded as Eq. (3),

$$z = \sum_{i=1}^4 \theta_{2,1,i} * \phi(\theta_{1,i,1} * \mu) \quad (4)$$

Fig. 4 Spatial attention gated path

where $*$ denotes the two-dimensional cross-correlation operator.

Finally, the channel attention gated path utilizes the Sigmoid function to normalize the attention scores and outputs the channel attention tensor $\mathbf{u} \in \mathbb{R}^C$, as shown in Eq. (5):

$$\mathbf{u} = \sigma(\mathbf{z}) \quad (5)$$

3.3 Spatial attention gated path

Fig. 4 plots the spatial attention gated path.

Firstly, the spatial attention gated path utilizes global average pooling to aggregate global channel information and generate spatial statistics $\mathbf{M} \in \mathbb{R}^{H \times W}$, as shown in Eq. (6):

$$\mu_{h,w} = \frac{1}{C} \sum_{k=1}^C x_{k,h,w} \quad (6)$$

Then, the spatial attention gated path utilizes a convolutional layer with the kernel size of 7×7 to establish spatial position correlations. Assuming that the weight of the convolutional layer is $\Theta \in \mathbb{R}^{7 \times 7}$. The process can be expressed as Eq. (7),

$$\mathbf{Z} = \Theta * \mathbf{M} \quad (7)$$

where $\mathbf{Z} \in \mathbb{R}^{H \times W}$.

Finally, the spatial attention gated path utilizes the Sigmoid function to normalize the attention scores and outputs the spatial attention tensor $\mathbf{V} \in \mathbb{R}^{H \times W}$, as shown in Eq. (8):

$$\mathbf{V} = \sigma(\mathbf{Z}) \quad (8)$$

To sum up, this section proposes the DBCSMA method, which establishes channel correlation and spatial correlation using two parallel gated paths, respectively. It then multiplicatively aggregates the outputs of the two gated paths and multiplies them by the input to recalibrate the channel information and spatial information of the input, thereby enhancing the channels of interest and spatial regions of interest. Next, Section 4 designs a series of experiments to evaluate and compare the proposed DBCSMA method with the current state-of-the-art channel-spatial mixed attention methods, thus demonstrating the superiority of the proposed DBCSMA method.

4 Experimental results and analysis

In this section, we performed a range of experiments to assess the efficacy of our DBC-SMA approach in comparison to the current state-of-the-art channel-spatial mixed attention methods across diverse tasks, datasets, and network architectures. All channel-spatial mixed attention methods were evaluated using identical experimental settings for fair benchmarking.

4.1 Implementation details

To evaluate and compare various channel-spatial mixed attention methods on CIFAR-100 [13] dataset, we utilized two commonly used CNNs, EfficientNet [17] and ResNet [18], as backbone models (all backbone network models only retained the last three spatial down-samplings). The network parameters were optimized using the Adam [19] optimizer with a mini-batch size of 128. The models were trained from scratch for 80 epochs, with an initial learning rate of $2e - 3$. During the training process, the learning rate scaling factor increased linearly from 0 to 1 for steps less than 390. For steps between 390 and 21450, the scaling factor decayed linearly with the step from 1 to 0.01. Finally, for steps greater than or equal to 21450, the scaling factor remained constant at 0.01.

Additionally, we evaluated and compared various channel-spatial mixed attention methods on PASCAL VOC [14] dataset using SSDlite [20] and RetinaNet [21]. For training SSDlite, the network was trained from scratch for 80 epochs using the Adam optimizer with a mini-batch size of 128. The initial learning rate was set to $4e - 3$. The learning rate scaling factor increased linearly from 0 to 1 for steps less than 133, decayed linearly with the step from 1 to 0.01 for steps between 133 and 7315, and remained constant at 0.01 for steps greater than or equal to 7315. For training RetinaNet, the network was trained from scratch for 40 epochs using the Adam optimizer with a mini-batch size of 32. The initial learning rate was set to $2e - 3$. The learning rate scaling factor increased linearly from 0 to 1 for steps less than 534, decayed linearly with the step from 1 to 0.01 for steps between 534 and 14418, and remained constant at 0.01 for steps greater than or equal to 14418.

For initialization, the weights in each convolution layer and fully connected layer were initialized with the Xavier truncated normal distribution [22], while biases if used were initialized with zeros. In each batch normalization layer [23], the scaling parameter (γ) was initialized with ones, while the shift parameter (β) if used was initialized with zeros.

All performance benchmark tasks were conducted on a single computer with the following specifications: CPU Intel(R) Core(TM) i7-11700K, GPU Nvidia RTX A5000, RAM 32GB, Python 3.11.7, PyTorch [24] 2.2.1, Cudatoolkit 12.1 and Cudnn 8.9.6. All time benchmark tasks were conducted on another single computer with the following specifications: CPU Intel(R) Xeon(R) E-2136, GPU Nvidia Quadro RTX 4000, RAM 16GB, Python 3.12.2, PyTorch 2.2.2, Cudatoolkit 12.1 and Cudnn 8.9.6.

4.2 Image classification results and analysis

To assess and contrast the performance of various mixed-attention methods, we initially conducted experiments using the CIFAR-100 dataset, which consists of color natural images with dimensions of 32×32 pixels. The CIFAR-100 dataset comprises 100 categories, each containing 600 images. While training on CIFAR-100, we employed standard data augmentation techniques such as random color space transformations and horizontal flips. We adopted the

Table 2 Comparison results between different channel-spatial mixed attention methods on CIFAR-100

Model	Top-1 err. (%) ↓	Top-5 err. (%) ↓	Param. (<i>M</i>) ↓
EfficientNet-B0 [17] + None	32.52	10.47	3.499
+ CBAM [4]	35.61	11.96	4.128
+ BAM [5]	32.43	10.06	4.705
+ FAB [8]	33.36	11.11	4.136
+ GALA [9]	62.25	32.47	4.458
+ TA [10]	54.96	26.93	3.504
+ CA [6]	33.31	11.06	4.459
+ SA [11]	32.61	10.37	3.506
+ SimAM [12]	32.14	10.87	3.499
+ EMA [7]	33.35	11.16	8.209
+ DBCSMA (Ours)	32.02	<u>10.20</u>	<u>3.501</u>
ResNet50 [18]	34.38	11.39	5.995
+ CBAM [4]	34.53	12.12	7.254
+ BAM [5]	33.24	10.69	9.323
+ FAB [8]	35.43	12.43	7.261
+ GALA [9]	61.95	32.75	7.897
+ TA [10]	36.01	12.83	6.000
+ CA [6]	34.49	11.60	7.898
+ SA [11]	33.85	11.54	5.998
+ SimAM [12]	33.61	11.55	5.995
+ EMA [7]	<u>33.37</u>	11.38	6.785
+ DBCSMA (Ours)	33.86	<u>11.33</u>	<u>5.997</u>

Note 1: None means the SE block used in the original backbone network is removed.

Note 2: The **bold** denotes the best result under the same backbone network.

Note 3: The underline denotes the second best result under the same backbone network

top-1 and top-5 error rates as our evaluation metrics. The experimental results were listed in Table 2.

To investigate the incorporation of various channel-spatial mixed attention mechanisms into contemporary network architectures, we chose EfficientNet-B0 and ResNet50 as our baseline models. When EfficientNet-B0 served as the baseline network, the original SE module was replaced by various channel-spatial mixed attention methods. When ResNet50 was used as the baseline network, the positions where different channel-spatial mixed attention methods were inserted followed those of the SE module in SENet [25].

Optimal performance analysis. As shown in Table 2, it could be observed that when the baseline network was EfficientNet-B0, DBCSMA achieved the lowest top-1 error rate of 32.02% and the second-lowest top-5 error rate of 10.20%. When the baseline network was ResNet50, DBCSMA achieved higher top-1 error rate compared to BAM, EMA, SimAM and SA, and higher top-5 error rate compared to BAM.

Robustness analysis. Fig. 5 further organized data from Table 2 to analyze the robustness of different channel-spatial mixed attention methods across different image classification networks. As shown in Fig. 5, it could be observed that integrating BAM, SimAM or DBCSMA into the baseline networks EfficientNet-B0 and ResNet50 improved the performance

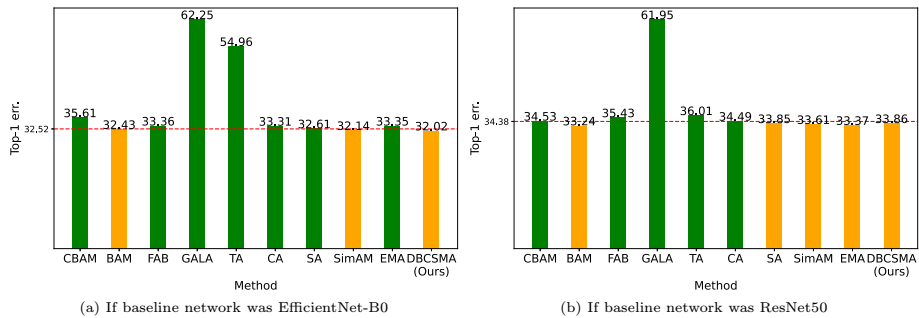


Fig. 5 Robustness analysis of different channel-spatial mixed attention methods across different image classification networks. Note: If the color of the bar chart corresponding to the same method in Figs. (a) and (b) is the same, it indicates that the method is robust

of the networks on CIFAR-100. This indicated that BAM, SimAM and DBCSMA exhibited greater robustness across different image classification networks.

4.3 Object detection results and analysis

Subsequently, we conducted experiments using the PASCAL VOC 2007/2012 dataset, which includes 20 classes. The PASCAL VOC 2012 trainval dataset contains 11530 images, each annotated with 27450 Region of Interest (ROI) objects. For comparison purposes, we trained all detectors on the VOC 2012 trainval dataset and assessed their performance on the VOC 2007 test dataset. During training, we utilized standard data augmentation techniques, such as random color space transformations and horizontal flips. The evaluation criteria were supplied by the COCO API. The experimental results were listed in Table 3.

Optimal performance analysis. As shown in Table 3, it could be observed that when the detector was SSDlite320, integrating DBCSMA achieves the highest mAP@50 of 37.3%, the highest mAP of 20.0%, and the second-highest mAP@75 of 19.0%. When the detector was RetinaNet50, integrating CA achieved the highest detection accuracy, while integrating DBCSMA achieved the second-highest detection accuracy.

Robustness analysis. Fig. 6 further organized the experimental results from Table 3 to analyze the robustness of different channel-spatial mixed attention methods across different detectors. From Fig. 6, it could be observed that integrating DBCSMA into both SSDlite320 and RetinaNet50 detectors improved their performance. This indicated that DBCSMA exhibited greater robustness across different detectors.

Visual analysis. Fig. 7 displays heatmaps based on Grad-CAM++ [26]. A random image was chosen, and backpropagation was applied to the highest-scoring bounding box to provide the gradients required by Grad-CAM++, enabling the generation of heatmaps. In the comparative analysis of Figs. 7b and 7c, both mainly focus on humans, with the highest heat concentrated in the head or central regions. Notably, Fig. 7c emphasizes the central area more effectively than Fig. 7b. In the comparative analysis of Figs. 7e and 7f, both mainly focus is on bicycles. Notably, Fig. 7f captures a more complete outline of the bicycle compared to Fig. 7e. In the comparative analysis of Figs. 7h and 7i, both mainly focus on cars. Notably, Fig. 7i reveals more details of the car's outline than Fig. 7h, e.g., the mirror.

Table 3 Comparison results between different channel-spatial mixed attention methods on PASCAL VOC 2007test

Model	mAP@50 (%) $\uparrow$	mAP (%) $\uparrow$	mAP@75 (%) $\uparrow$	Param. (M) $\downarrow$
SSDlite320 [20] + None	36.0	19.2	18.4	1.650
+ CBAM [4]	33.3	17.4	16.4	2.467
+ BAM [5]	<u>36.6</u>	<u>19.9</u>	19.5	4.728
+ FAB [8]	35.2	18.9	18.3	2.470
+ GALA [9]	23.6	11.7	10.5	2.881
+ TA [10]	20.6	9.9	8.7	1.652
+ CA [6]	32.6	17.4	16.6	2.882
+ SA [11]	36.3	19.2	18.3	1.652
+ SimAM [12]	34.4	18.4	17.8	1.650
+ EMA [7]	35.4	18.6	17.4	2.671
+ DBCSMA (Ours)	37.3	20.0	<u>19.0</u>	<u>1.651</u>
RetinaNet50 [21]	34.3	17.6	16.1	14.542
+ CBAM [4]	37.9	19.8	19.0	15.801
+ BAM [5]	27.6	13.9	12.3	17.870
+ FAB [8]	30.5	14.8	12.7	15.808
+ GALA [9]	16.0	6.6	4.2	16.444
+ TA [10]	31.9	15.5	13.3	14.547
+ CA [6]	40.3	21.4	20.5	16.445
+ SA [11]	36.1	18.7	17.3	14.545
+ SimAM [12]	34.3	17.8	16.5	14.542
+ EMA [7]	36.9	19.1	17.9	15.332
+ DBCSMA (Ours)	<u>38.7</u>	<u>20.2</u>	<u>18.9</u>	<u>14.544</u>

Note 1: None refers to Note 1 in Table 2.

Note 2: The **bold** refers to Note 2 in Table 2.

Note 3: The underline refers to Note 3 in Table 2

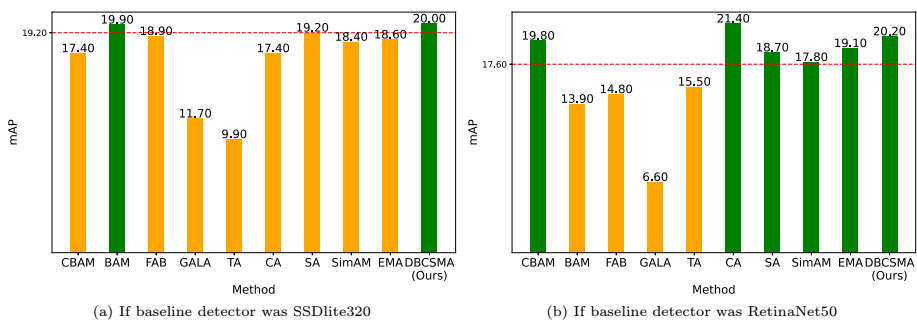


Fig. 6 Robustness analysis of different channel-spatial mixed attention methods across different detectors. Note: Refer to the Note in Fig. 5

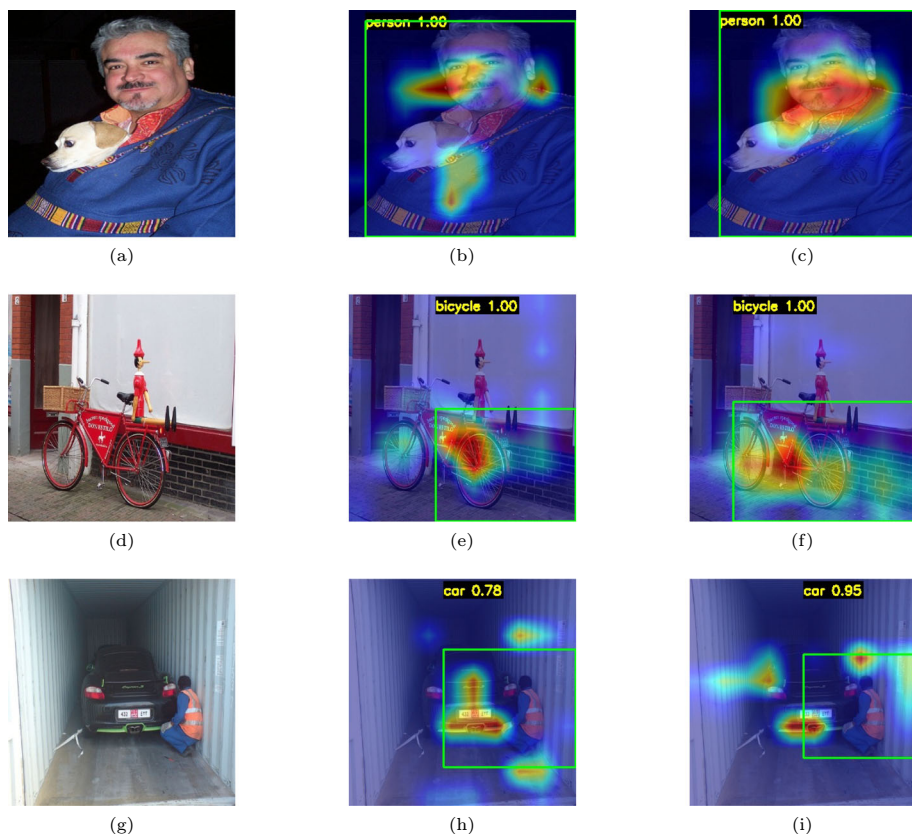


Fig. 7 Visual analysis. Note: The first column denotes the original images, the second column denotes the heatmaps of “SSDlite320 + None”, and the third column denotes the heatmaps of “SSDlite320 + DBCSMA”

4.4 Complexity results and analysis

Table 4 listed the complexity of different channel-spatial mixed attention methods. The inference speed was measured as follows: The input size was $X \in \mathbb{R}^{32 \times 1024 \times 56 \times 56}$ (32 for batch size, 1024 for the number of channels, 56 for height, and 56 for width), and the measurements were repeated 300 times.

Computational complexity analysis. As shown in Table 4, it could be observed that among all the compared channel-spatial mixed attention methods, DBCSMA achieved the fastest inference speed on GPU and the second-fastest inference speed on CPU.

Parameter complexity analysis. As shown in Table 4, except for the parameterless SimAM, DBCSMA exhibited lower parameter complexity compared to other channel-spatial mixed attention methods.

4.5 Ablation analysis

In this section, ablation studies were conducted to analyze the impact of different structural designs on DBCSMA. To evaluate the effect of different aggregation methods of channel

Optimally increasing upwards	SimAM	DBCSMA (Ours)	DBCSMA (Ours)	BAM	DBCSMA (Ours)	CA
	DBCSMA (Ours)	CA	SimAM	EMA	BAM	DBCSMA (Ours)
	TA	SA	BAM	SimAM	SA	CBAM
	SA	SimAM	SA	SA	FAB	EMA
	FAB	CBAM	CA	DBCSMA (Ours)	EMA	SA
	CBAM	TA	EMA	CA	SimAM	SimAM
	EMA	FAB	FAB	CBAM	CBAM	TA
	CA	GALA	CBAM	FAB	CA	FAB
	GALA	BAM	TA	TA	GALA	BAM
	BAM	EMA	GALA	GALA	TA	GALA
	Param.	FPS (GPU)	Top-1 err. (EfficientNet-B0)	Top-1 err. (ResNet50)	mAP (SSDLite320)	mAP (RetinaNet50)

Fig. 8 Ranking of different channel-spatial mixed attention methods using different evaluated metrics

Table 4 Inference speed and parameters of different channel-spatial mixed attention methods

Method	CPU (FPS) ↑	GPU (FPS) ↑	Param. ↓
CBAM[4]	3.03	58.92	524386
BAM[5]	0.52	16.67	1968386
FAB[8]	0.97	34.57	524288
GALA[9]	1.00	31.82	788736
TA[10]	1.64	49.59	294
CA[6]	5.25	<u>109.28</u>	786944
SA[11]	1.96	65.28	768
SimAM[12]	1.63	59.84	0
EMA[7]	0.55	12.09	655872
DBCSMA (Ours)	<u>4.72</u>	148.11	<u>105</u>

Note 1: The **bold** denotes the best result.

Note 2: The underline denotes the second best result.

Note 3: FPS stands for Frames Per Second

Table 5 Influence of different aggregation methods of channel attention gated path and spatial attention gated path on DBCSMA performance

Method	Top-1 err. ↓	Top-5 err. ↓
Avg.	32.98	10.35
Random Avg.	32.88	10.59
Series	32.45	10.45
Mul.	32.02	10.20

attention gated path and spatial attention gated path on the performance of DBCSMA, ablation experiments were performed using EfficientNet-B0 on the CIFAR-100 dataset in this section. The experimental settings were the same as those in Sec. 4.1.

Table 5 listed the corresponding experimental results. The descriptions of each method in Table 5 were as follows. Avg. denotes $X \odot (CAGP(X) + SAGP(X))/2$. Random avg. denotes $X \odot (\lambda CAGP(X) + (1 - \lambda)SAGP(X))$. Series denotes $(X \odot CAGP(X)) \odot SAGP(X \odot CAGP(X))$. Mul. denotes $X \odot (CAGP(X) \odot SAGP(X))$. Experimental results showed that multiplicative aggregation achieved the best results.

4.6 Discussion

Fig. 8 illustrates the ranking of different channel-spatial mixed attention methods based on different evaluated metrics. This ranking is derived from a comprehensive analysis of the experimental results in Sec. 4.

According to Fig. 8, SimAM had less parameters than DBCSMA, but it performed worse than DBCSMA on the other five evaluated metrics. BAM achieved better top-1 error rate metric than DBCSMA on ResNet50, but it performed worse than DBCSMA on the other five evaluated metrics. CA achieved better mAP metric than DBCSMA on RetinaNet50, but it performed worse than DBCSMA on the other five evaluated metrics. Further analysis reveals that DBCSMA is the optimal channel-spatial mixed attention method under the same order of magnitude of parameter complexity and computational complexity.

In the future, DBCSMA can be applied to more fields, *e.g.*, lightweight network architecture design [20, 27], representation learning [28], graph neural networks [29, 30], and recommendation systems [31–33].

5 Conclusion

This paper proposes DBCSMA, which is aimed at simultaneously focusing on both the objects of interest and spatial regions of interest. Specifically, DBCSMA consists of two parallel gated paths: the channel attention gated path and the spatial attention gated path. The channel attention gated path captures channel relationships using the Inverted Bottleneck and outputs a normalized channel attention tensor. The spatial attention gated path establishes spatial relationships using the convolutional operation and outputs a normalized spatial attention tensor. The two parallel gated paths are multiplicatively aggregated to produce a mixed attention tensor. By means of multiplying the mixed attention tensor with the input, the input channel and spatial position information are recalibrated. Experimental results demonstrate that DBCSMA is the optimal channel-spatial mixed attention method under the same order of magnitude of parameter complexity and computational complexity.

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Data Availability The datasets during the current study are open source and available in <http://www.cs.toronto.edu/~protect/unhbox/voidb@x\penalty@\M\kriz/cifar.html> and <http://host.robots.ox.ac.uk/pascal/VOC/voc2012/index.html> repositories, respectively.

Declarations

Conflicts of Interest The authors declare no conflict of interest.

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